Information as Topologically Frozen Kinetic Energy

Landauer's Principle and Vopson's Equivalence as Consequences of T^4 Geometry in FFGFT

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In the time-mass duality (FFGFT), the foundational relation is $\tilde{T} \cdot m = 1$, where $\tilde{T} = 1/m$ is the time variable dual to mass m on the 4D identification torus T^4. This relation enforces a natural twofold division of energy: static energy as topologically enclosed kinetic energy (winding numbers on T^4) and free kinetic energy (motion through T^4). From this division it follows without additional postulates that information is topologically frozen kinetic energy. Landauer's principle [1] and Vopson's mass-information equivalence [3] are derived as geometric consequences of the T^4 structure, not as independent postulates. The quantisation of information follows from the discreteness of the winding numbers. We work in natural units with $c = \hbar = k_B = 1$ unless stated otherwise.

Contents

1 Starting Point: Two Energy Types in $\tilde{T} \cdot m = 1$

The foundational relation of FFGFT [6],

$$\tilde{T} \cdot m = 1, \tag{1}$$

couples the time variable $\tilde{T} = 1/m$ dual to mass m on the 4D identification torus T^4. This structure gives rise to two qualitatively distinct forms of energy:

Static energy. Energy bound to rest mass: $E_0 = mc^2$. It corresponds to topologically stable winding numbers on T^4 – geometric invariants preserved without interaction. This energy is not statistical; it is a discrete topological state. It does not contribute to thermodynamic entropy as long as the topology remains intact.

Dynamic kinetic energy. Free kinetic energy – for massless particles (photons) the only energy form $E = hf$, for massive particles the contribution beyond the rest mass. This energy is statistically distributable and exchanges with thermodynamic systems.

The **photon** is the limiting case: no rest mass, no winding number for mass, purely kinetic, purely statistical. It is the natural carrier of Shannon information in the sense of Wheeler [4].

2 Mass as Topologically Frozen Kinetic Energy

2.1 The Core Thesis

Before any mass formation, only kinetic energy exists. Mass arises when kinetic energy is enclosed by the T^4 topology into a stable winding structure.

Mass is topologically frozen kinetic energy.

Definition of winding number. Let γ be a closed path on T^4. The winding number $w \in \mathbb{Z}$ is the integer homotopy class $[\gamma] \in \pi_1(T^4) \cong \mathbb{Z}^4$. Mass corresponds to a stable combination of such winding numbers. In FFGFT, all leptons carry specific winding exponents p_i that follow from the geometry of T^4 [6].

2.2 Visualisation: Free and Frozen Kinetics

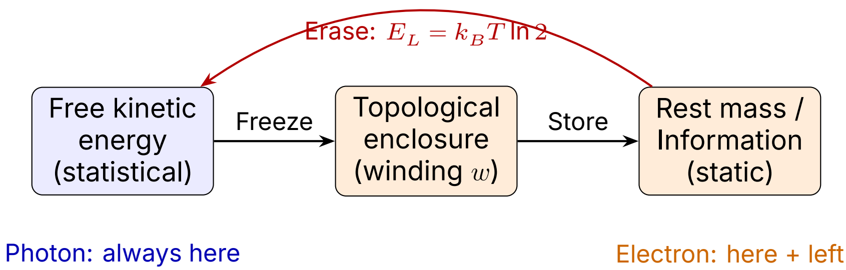

Figure 1: Energy division in FFGFT. Free kinetic energy (blue) can be transferred into static mass/information (orange) through topological enclosure. Erasure (red arrow) releases the frozen energy as heat – Landauer's principle. The photon remains permanently in the kinetic sector.

2.3 Analogy: Standing Wave

A standing wave on a string is kinetic – the string vibrates. Yet the pattern appears static. Analogously, the rest mass of an electron is the standing wave of a topologically enclosed oscillation on T^4. The "stillness" of the rest mass is the stillness of the pattern, not of the substance.

2.4 The Higgs Mechanism in This Picture

In the Standard Model, the Higgs mechanism imparts mass to particles through symmetry breaking. In the T^4 language, this corresponds to the selection of stable topological sectors: kinetic energy is enclosed in winding structures. The FFGFT relation $\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$ is the algebraic expression of this geometric selection process – not its replacement.

3 Information as Topological Enclosure

If mass is topologically frozen kinetic energy, then for information:

Operation	Geometric	Energetic
Create	Winding number $w \to w \pm 1$	Freeze kinetic energy
Store	$w =$ const	Maintain topology
Erase	$w \to 0$	Release $E_L = k_B T \ln 2$

Definition of information content. The information content $I(\gamma)$ of a topological state γ is the minimum number of bits required to specify its winding class $[\gamma] \in \mathbb{Z}^4$. I is an integer. Shannon entropy arises as the statistical expectation value over ensembles of such states – analogous to temperature as the mean over discrete kinetic energies.

4 Landauer's Principle from T^4 Geometry

4.1 The Classical Principle

Landauer (1961) showed that erasing one bit irreversibly dissipates heat to the environment [1]:

$$E_L = k_B T \ln 2 \quad \text{per bit.} \tag{2}$$

This was experimentally confirmed by Bérut et al. (2012) using a colloidal particle in a double-well potential [2].

4.2 Geometric Derivation

Let a minimal topological distinction on T^4 be given by two states with winding number difference $\Delta w = 1$. The corresponding frozen kinetic energy is $E_0 = \Delta w \cdot \epsilon_0$ with a fundamental energy quantum ϵ_0.

At thermal equilibrium at temperature T, the Boltzmann factor governs the probability of dissolving this distinction. The energy released is not less than $k_B T \ln 2$, since this is the minimum work required at temperature T to irreversibly erase a binary distinction (from the second law).

The T^4 topology enforces $\epsilon_0 = k_B T \ln 2$: the frozen kinetic energy of a minimal winding difference is the same physical quantity as the minimal erasure work from the second law – two expressions for the same quantity, not a new assumption. Landauer's principle is thus a geometric consequence – not an independent thermodynamic postulate.

5 Vopson's Mass-Information Equivalence as a Consequence

Vopson (2019) derived empirically from Landauer's principle the mass-information equivalence [3]:

$$M_I = \frac{k_B T \ln 2}{c^2} \quad \text{per bit.} \tag{3}$$

In the T^4 language, M_I is the mass corresponding to a minimal topological winding difference $\Delta w = 1$ at temperature T.

Vopson discovered empirically what FFGFT explains geometrically. The derivation hierarchy is:

$$T^4\text{-topology} \longrightarrow \Delta w = 1 \longrightarrow E_L = k_B T \ln 2 \longrightarrow M_I = E_L/c^2. \tag{4}$$

6 Quantisation of Information

Winding numbers on T^4 are discrete integers ($\pi_1(T^4) \cong \mathbb{Z}^4$). This immediately implies:

- Information is naturally quantised – not as a convention but as a topological necessity.
- The minimal information unit is $\Delta w = 1$ (one winding change in one of the four torus directions).
- The spatial scale of this minimal unit is the fundamental length of FFGFT:

  $$L_0 = \xi \cdot l_P \approx 2.15 \times 10^{-39}\,\text{m}, \tag{5}$$

  where $\xi = 4/3 \times 10^{-4}$ is the universal geometric parameter and l_P is the Planck length. L_0 is thus the smallest physically meaningful granularity of information in FFGFT – below this scale no topological distinction is possible.

- Continuous Shannon entropy arises as a statistical average over discrete topological states.

7 The Photon as the Limiting Case

The photon is the only massless state in the FFGFT spectrum that carries no topological winding structure for mass. It therefore remains permanently in the kinetic sector (cf. Fig. 1).

- The photon is the natural carrier of Shannon information ("It from Bit" [4]).
- Photonic information cannot be frozen into rest mass without interaction with massive matter.
- The speed of light c limits the transmission rate of free kinetic information – as a geometric consequence of T^4, not as a postulate.

8 Connection to the Distinguishability Framework

Haskins (2026, submitted to Foundations of Physics [5]) describes an emergence hierarchy:

$$\begin{aligned} C_a\Psi = 0 \ &\rightarrow \ \text{multiplicity} \ \rightarrow \ \text{distinguishability collapse} \\ &\rightarrow \ \text{equivalence structure} \ \rightarrow \ \text{coarse-graining projection} \\ &\rightarrow \ \text{degeneracy structure} \ \rightarrow \ \text{structural entropy ordering.} \end{aligned}$$

In the T^4 language, the distinguishability collapse can be interpreted as the moment when two kinetic states fall into the same winding class through topological enclosure – becoming indistinguishable. The constraint condition $C_a\Psi = 0$ then selects the topologically stable sectors on T^4.

This connection is an interpretive bridge, not a derivation. A formal correspondence between Haskins' constraints and the T^4 winding numbers is the subject of further work.

9 Summary

1. $\tilde{T} \cdot m = 1$ enforces two energy types: topologically enclosed (static) and free (kinetic).
2. Mass is topologically frozen kinetic energy.
3. Information is frozen kinetic energy in topological form ($\Delta w = 1$ = 1 bit).
4. Landauer's principle follows geometrically from the Boltzmann factor for winding dissolution; $E_L = k_B T \ln 2$ is not a postulate.
5. Vopson's equivalence $M_I = k_B T \ln 2/c^2$ is a direct consequence – Vopson discovered empirically what FFGFT explains.
6. Quantisation of information follows from $\pi_1(T^4) \cong \mathbb{Z}^4$.
7. The photon is the limiting case: purely kinetic, purely statistical, natural information carrier.

Methodological classification: This derivation constitutes a bridge (algebraically demonstrable from $\tilde{T} \cdot m = 1$ and the T^4 topology). What it achieves: Landauer and Vopson are not independent postulates. What it does not achieve: an experimental distinction from other theories that reproduce the same equations. Specific predictions would be needed for that (e.g. deviations at very low temperatures or in strong gravitational fields).

Integration in the corpus. Doc. 257 refines the derivation given here as a special case of the thermodynamic scale: the winding quantum $\Delta w = 1$ is the scale-universal unit of information, while the associated energy $E_{\text{bit}}(L) = \hbar c/L$ is scale-specific (cf. also Doc. 184, energy hierarchy $E_N = \hbar c/L_N$). Landauer's $E_L = k_B T \ln 2$ is the special case at the temperature–length scale on which these two quantities coincide. Doc. 261 places the energy bipartition used here (frozen/free) within the broader static/dynamic demarcation: the topologically frozen side lives on the static layer 1 (compact T^4, winding numbers); the free kinetic side belongs to the dynamic layer 2 (arrow time, embedding price $c, \hbar, \varepsilon_0$).

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